

## AL HOCEIMA CHERIF AL IDRISI, Morocco

WMO: 601070

Lat: 35.1803N

Lon: 3.8437W

Elev: 14

StdP: 101.16

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 5.8        | 6.9 | -1.7                          | 3.3 | 10.1 | 0.8 | 4.0 | 11.4 | 10.0                  | 14.2 | 8.4 | 14.9 | 1.8                   | 180  | 0.340 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 7.1                    | 31.2            | 20.9 | 29.8 | 21.3 | 28.3 | 21.6 | 25.0                | 27.6 | 24.2 | 26.9 | 23.6 | 26.2 | 4.1                  | 20   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 24.1                            | 19.0 | 26.8 | 23.2 | 18.0 | 25.9 | 22.8 | 17.5 | 25.6 | 75.8          | 27.7 | 73.0 | 26.9 | 70.2 | 26.2 | 29.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 8.3               | 7.2  | 6.3 | 3.7                        | 34.8 | 1.8                | 1.9 | 2.4                                                | 36.2 | 1.3        | 37.3 | 0.3        | 38.4 | -1.1       | 39.8 |    |    |
|                   |      |     | 1.9                        | 25.8 | 1.9                | 1.5 | 0.6                                                | 26.9 | -0.5       | 27.7 | -1.6       | 28.5 | -2.9       | 29.6 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 18.4   | 12.9  | 13.4  | 14.9  | 16.3  | 19.0  | 22.0  | 24.2  | 24.7  | 22.5  | 19.9  | 16.2  | 13.9 |
|                                                                                  | DBStd       | 4.55   | 2.00  | 1.97  | 1.85  | 1.78  | 1.97  | 1.78  | 1.88  | 1.83  | 1.91  | 1.97  | 2.37  | 1.95 |
|                                                                                  | HDD10.0     | 4      | 2     | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1    |
|                                                                                  | HDD18.3     | 708    | 168   | 137   | 108   | 64    | 14    | 0     | 0     | 0     | 7     | 71    | 137   |      |
|                                                                                  | CDD10.0     | 3060   | 93    | 98    | 152   | 188   | 279   | 361   | 440   | 457   | 375   | 308   | 187   | 122  |
|                                                                                  | CDD18.3     | 721    | 0     | 0     | 1     | 3     | 36    | 111   | 182   | 198   | 125   | 57    | 8     | 1    |
|                                                                                  | CDH23.3     | 3993   | 1     | 0     | 9     | 12    | 101   | 466   | 1207  | 1429  | 595   | 151   | 22    | 1    |
|                                                                                  | CDH26.7     | 813    | 0     | 0     | 1     | 1     | 10    | 74    | 280   | 346   | 87    | 13    | 2     | 0    |
| Wind                                                                             | WSAvg       | 2.5    | 2.3   | 2.6   | 2.7   | 2.8   | 2.7   | 2.9   | 2.7   | 2.6   | 2.4   | 2.1   | 2.3   | 2.2  |
| Precipitation                                                                    | PrecAvg     | 368    | 45    | 46    | 44    | 38    | 23    | 10    | 2     | 6     | 17    | 45    | 53    | 48   |
|                                                                                  | PrecMax     | 657    | 144   | 215   | 112   | 151   | 77    | 120   | 9     | 34    | 46    | 351   | 208   | 155  |
|                                                                                  | PrecMin     | 214    | 0     | 0     | 0     | 3     | 1     | 0     | 0     | 1     | 1     | 2     | 0     | 3    |
|                                                                                  | PrecStd     | 115    | 41    | 43    | 30    | 33    | 22    | 25    | 2     | 8     | 13    | 67    | 47    | 39   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 21.2  | 21.8  | 24.6  | 24.9  | 28.0  | 30.9  | 34.0  | 33.8  | 30.7  | 28.7  | 25.9  | 22.2 |
|                                                                                  |             | MCWB   | 14.4  | 14.2  | 15.4  | 15.6  | 17.9  | 19.5  | 20.4  | 21.4  | 22.0  | 20.5  | 16.6  | 15.0 |
|                                                                                  | 2%          | DB     | 19.6  | 20.1  | 21.8  | 23.0  | 25.8  | 28.7  | 31.4  | 31.2  | 28.9  | 26.2  | 23.2  | 20.2 |
|                                                                                  |             | MCWB   | 14.0  | 14.0  | 15.0  | 15.9  | 18.1  | 20.0  | 20.3  | 21.5  | 22.4  | 20.5  | 17.2  | 14.8 |
|                                                                                  | 5%          | DB     | 18.2  | 18.9  | 20.1  | 21.6  | 24.1  | 27.1  | 29.4  | 29.9  | 27.4  | 25.0  | 21.9  | 19.1 |
|                                                                                  |             | MCWB   | 13.6  | 13.6  | 14.5  | 15.5  | 17.9  | 20.0  | 21.2  | 21.8  | 22.1  | 20.1  | 17.0  | 14.6 |
|                                                                                  | 10%         | DB     | 17.1  | 17.9  | 19.0  | 20.2  | 23.0  | 25.9  | 28.0  | 28.4  | 26.2  | 23.9  | 20.8  | 18.1 |
|                                                                                  |             | MCWB   | 13.2  | 13.5  | 14.4  | 15.4  | 17.7  | 19.9  | 21.2  | 22.3  | 21.5  | 19.7  | 16.6  | 14.1 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 16.5  | 16.1  | 17.5  | 18.5  | 20.8  | 24.0  | 25.4  | 26.2  | 25.1  | 22.7  | 19.7  | 17.4 |
|                                                                                  |             | MCDB   | 19.1  | 18.9  | 20.7  | 21.7  | 24.3  | 27.2  | 28.4  | 28.8  | 27.7  | 25.6  | 22.4  | 19.2 |
|                                                                                  | 2%          | WB     | 15.2  | 15.2  | 16.5  | 17.5  | 20.0  | 22.7  | 24.5  | 25.3  | 24.1  | 21.8  | 18.7  | 16.4 |
|                                                                                  |             | MCDB   | 17.9  | 18.3  | 19.5  | 20.4  | 23.1  | 25.4  | 27.2  | 27.8  | 26.7  | 24.6  | 21.8  | 18.5 |
|                                                                                  | 5%          | WB     | 14.4  | 14.5  | 15.7  | 16.8  | 19.3  | 21.8  | 23.7  | 24.6  | 23.3  | 21.1  | 17.8  | 15.5 |
|                                                                                  |             | MCDB   | 17.3  | 17.6  | 18.6  | 19.7  | 22.4  | 24.6  | 26.5  | 27.2  | 25.9  | 23.7  | 20.8  | 18.1 |
|                                                                                  | 10%         | WB     | 13.6  | 13.9  | 15.1  | 16.2  | 18.6  | 21.2  | 23.1  | 23.9  | 22.5  | 20.4  | 17.1  | 14.7 |
|                                                                                  |             | MCDB   | 16.6  | 16.9  | 18.0  | 19.2  | 21.8  | 24.0  | 25.8  | 26.4  | 25.3  | 23.0  | 19.8  | 17.6 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.7   | 8.3   | 8.1   | 7.9   | 7.3   | 7.0   | 7.1   | 7.1   | 7.0   | 7.3   | 8.3   | 8.6  |
|                                                                                  |             | MCDBR  | 10.1  | 10.2  | 10.3  | 10.2  | 9.4   | 9.2   | 9.9   | 9.5   | 8.5   | 9.0   | 9.6   | 9.6  |
|                                                                                  |             | MCWBR  | 6.6   | 6.3   | 6.1   | 5.8   | 5.2   | 4.6   | 5.3   | 5.1   | 4.9   | 5.3   | 5.7   | 6.2  |
|                                                                                  | 5% WB       | MCDBR  | 8.8   | 8.5   | 8.3   | 7.9   | 7.6   | 6.9   | 6.5   | 6.8   | 6.9   | 7.5   | 7.9   | 8.7  |
|                                                                                  |             | MCWBR  | 6.3   | 5.8   | 5.7   | 5.3   | 4.7   | 4.0   | 4.4   | 4.5   | 4.6   | 5.0   | 5.2   | 6.2  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.330  | 0.364 | 0.424 | 0.431 | 0.462 | 0.500 | 0.535 | 0.524 | 0.476 | 0.427 | 0.372 | 0.339 |      |
|                                                                                  | taud        | 2.496  | 2.393 | 2.196 | 2.204 | 2.144 | 2.042 | 1.948 | 1.990 | 2.134 | 2.267 | 2.410 | 2.490 |      |
|                                                                                  | Ebn at Noon | 872    | 874   | 843   | 856   | 831   | 796   | 765   | 765   | 783   | 792   | 810   | 836   |      |
|                                                                                  | Edh at Noon | 84     | 104   | 137   | 143   | 154   | 170   | 185   | 175   | 144   | 116   | 91    | 80    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.80   | 3.62  | 4.75  | 5.79  | 6.68  | 7.50  | 7.37  | 6.53  | 5.28  | 3.95  | 2.93  | 2.51  |      |
|                                                                                  | RadStd      | 0.24   | 0.36  | 0.45  | 0.43  | 0.51  | 0.22  | 0.17  | 0.26  | 0.24  | 0.28  | 0.23  | 0.20  |      |

## Historical Trends

|                        |       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                    | (e)   | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| Station Only           | N/A   | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (9 neighbors) | +0.28 | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | +94     | +82     |         |

Nomenclature: See separate page